KENBRIGHT ACTUARIAL AND FINANCIAL SERVICES LIMITED

ORDINARY LIFE INSURANCE VALUATION

INTERNAL TECHNICAL PROCEDURES

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OUR REF

\\Kenbright\Internal Technical Procedures\Ordinary Life Insurance Valuation

DOCUMENT PURPOSE

This document has been produced as part of the internal technical procedures for KAFS and provides guidance on valuation processes for ordinary life insurance business

DOCUMENT OWNERSHIP

The owner of this document is Kenbright Actuarial and Financial Services Limited

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ABBREVIATIONS

NAME	DEFINITION
CSM	Contractual Service Margin
EPV	Expected Present Value
IRA	Insurance Regulatory Authority
KAFS	Kenbright Actuarial & Financial Services Limited
LC	Loss Component
LIC	Liability for Incurred Claims
LRC	Liability for Remaining Coverage
NSE	Nairobi Securities Exchange
OCR	Outstanding Claims Reserves
RA	Risk Adjustment
Valuation Date	The date at which liabilities of an insurer are determined, e.g. final day of a quarter or financial year

1 INTRODUCTION

Purpose

1.1. The IRA issues guidelines for insurers operating in Kenya on conducting life insurance valuations to ensure that policyholder liabilities are appropriately measured and reported.

1.2. The purpose of life insurance valuation is to establish the present value of insurers' obligations to policyholders, reflecting realistic assumptions about mortality, expenses, lapses, investment returns, and other key risks. This ensures that reported liabilities are prudent, reliable, and aligned with international actuarial standards.

1.3. This manual explains the ordinary life insurance valuation framework in Kenya as regulated by the IRA, outlining the methodology, assumptions, and regulatory requirements applicable to life insurers.

Key Concepts

1.4. **Valuation of Liabilities**: This refers to the process of determining the present value of future policyholder benefits and related obligations, using prudent actuarial assumptions on mortality, interest, expenses, and persistency.

1.5. **Assumptions:** Actuarial assumptions include mortality/morbidity, interest rates (discount rate), expense inflation, and policyholder behavior (e.g., lapses and surrenders). These must be reasonable, supportable, and, in some cases, prescribed by the IRA.

1.6. **Fulfilment cash flows**: EPV of the future cash outflows minus the EPV of the future cash inflows that arise as the entity fulfils insurance contracts, including risk adjustment for non-financial risk.

1.7. **Risk Adjustment for non-financial risk:** The compensation an entity requires for bearing the uncertainty about the amount and timing of the cash flows that arise from non-financial risk as the entity fulfils insurance contracts.

1.8. **Contractual Service Margin**: A component of the carrying amount of the asset or liability for a group of insurance contracts representing the unearned profit the entity will recognize as it provides insurance contract services under the insurance contracts in the group.

2 ORDINARY LIFE INSURANCE PRODUCTS

Overview

2.1. Ordinary life insurance products provide long-term **protection** and **savings** solutions to policyholders. The products are characterized by different structures, for example, pure protection contracts, such as term assurance, savings-oriented contracts, such as endowments and annuities.

2.2. The valuation of such products is a critical actuarial exercise, as it ensures that insurers hold adequate reserves to meet their policyholder obligations while also ensuring long-term financial soundness.

2.3. These products can generally be categorized into two types; **with-profit** and **without-profit** policies.

a. With-Profit Policies

2.4. With-profit policies entitle policyholders to a share in the insurer's surplus through the allocation of bonuses.

2.5. These bonuses are not guaranteed in advance but are declared periodically depending on the insurer's investment performance, expense experience, and overall financial results.

2.6. Policyholders receive guaranteed benefits plus bonuses.

2.7. The two most common types of bonuses are Reversionary and Terminal Bonuses.

i. Reversionary Bonus (Simple or Compound)

2.8. Reversionary bonus is a regular addition made to the sum assured during the policy term. Once declared, it becomes a guaranteed part of the policy benefit and cannot be withdrawn.

2.9. Simple Reversionary Bonus is calculated as a percentage of the original sum assured and added each year.

2.10. Compound Reversionary Bonus is calculated as a percentage of the sum assured plus previously declared bonuses, which allows the benefit to grow at a compounded rate. Example;

> *"If a policyholder has a Whole Life Assurance policy with a sum assured of **KES 1,000,000**, and the insurer declares a **2% reversionary bonus**, the benefit increases by **KES 20,000** in that year. This additional amount becomes guaranteed and will be paid at maturity or on death, along with the original sum assured".*

ii. Terminal Bonus (Final Bonus)

2.12. Terminal bonus is a one-off addition made to the policy benefits at the end of the policy term, either on maturity or on the death of the policyholder.

2.13. Unlike the reversionary bonus, the terminal bonus is not guaranteed until it is declared, and its amount depends heavily on the insurer's long-term investment performance and the overall experience of the with-profits fund. Example;

> *"At the maturity of the same Whole Life Assurance policy with a sum assured of **KES 1,000,000**, the insurer may declare a terminal bonus of **KES 50,000**. The final benefit would therefore be; (Sum Assured + Accumulated Reversionary Bonuses + Terminal Bonus)".*

b. Without-Profit Policies

2.14. Without-profit policies (also known as non-participating policies) provide policyholders with benefits that are contractually guaranteed at inception. Policyholders do not share in the insurer's surplus.

2.15. Policyholders receive a fixed sum assured or guaranteed maturity value.

2.16. Premiums are determined to cover expected claims, expenses, and a margin for profit, without any bonus additions.

Types of Ordinary Life Products

ii. **Whole Life Assurance-** These products provide coverage for the entire life of the insured. The benefit (sum assured) is payable on death, whenever it occurs. Premiums are usually level and payable throughout life (or for a specified period like 30 years).

iii. **Term Assurance-** These provide coverage for a fixed period (e.g., 10, 20, or 30 years). The sum assured is payable only if the insured dies within the term.

iv. **Endowment Assurance-** These products pay the sum assured either on death during the term or survival to the end of the term. They combine the aspect of savings and protection.

v. **Anticipated Endowment Assurance**- This is a type of endowment where portions of the sum assured are paid at intervals during the policy term, with the balance paid at maturity or on earlier death.

vi. **Annuities** (Immediate and Deferred)- These provide regular income payments to the policyholder either immediately (immediate annuity) or after a deferment period (deferred annuity).

vii. **Riders / Supplementary Benefits**- Riders are extra benefits attached to main life policies. They are usually short-term in nature but attached to long-term contracts. Some common examples include;

i. Accidental Death Benefit (ADB)- This rider provides an extra benefit if the policyholder dies as a result of an accident. It is usually paid in addition to the basic sum assured under the main policy

ii. Waiver of Premium- Under this rider, future premiums are waived if the policyholder becomes totally disabled or critically ill and is unable to work. The policy remains in force, and benefits are unaffected, even though no further premiums are paid.

iii. Disability rider- This rider provide financial protection if the policyholder becomes permanently or temporarily disabled.

iv. Critical Illness riders- This rider pays a lump sum if the policyholder is diagnosed with a specified critical illness (e.g., cancer, heart attack, stroke). The payout helps cover medical expenses and loss of income.

2.17. Riders increase the total premium payable by the policyholder. The base premium relates to the main policy, while each rider attracts an additional charge depending on the level of cover and the risk involved.

2.18. For example, if a person buys a Whole Life Assurance policy and then adds an Accidental Death Benefit, the total benefit payable would be higher if death occurs due to an accident, since the rider provides additional cover on top of the base policy.

3 VALUATION OF LIABILITIES

3.11. With the global adoption and implementation of the IFRS 17 Standard, ordinary life insurance valuation procedures outlined below have been based on the Standard.

3.12. Insurance liabilities include both the insurer's Liability for Remaining Coverage and the Liability for Incurred Claims.

3.13. Under the IFRS 17 Standard, these liabilities are as identified below.

Liability for Remaining Coverage (LRC)

3.14. The insurance liabilities for ordinary life business are determined as the Liability for Remaining Coverage**,** which represents the cost of running off the unexpired portion of the insurer's policies. These comprise of;

a. Fulfilment Cashflows

b. Contractual Service Margin

c. Risk Adjustment

EPV of Future Cashflows (Fulfilment Cashflows)

3.15. Fulfilment cashflows represents the expected present value of future cash flows (benefits and expenses less premiums) based on best estimate assumptions and discounted using the relevant risk-free interest rate.

i. **Cash outflows** include benefit payments (claims, maturities, surrenders), acquisition costs, ongoing servicing expenses, and any other expected payments under the contract.

ii. **Cash inflows** primarily include premiums receivable and recoveries (e.g., reinsurance).

iii. The estimation uses **best estimate assumptions** on mortality, morbidity, persistency, expenses, inflation, and lapses.

iv. Future cashflows are **discounted** using a **the NSE risk-free yield curve**.

v. The EPV of future cashflows represents the **probability-weighted** expectation of all future inflows and outflows, forming the base of the insurance contract liability.

Contractual Service Margin (CSM)

3.16. This is a component of the carrying amount of the asset or liability for a group of insurance contracts representing the unearned profit the entity will recognize as it provides services under the insurance contracts in the group.

3.17. The CSM is established at inception as the difference between the fulfilment cashflows and the premiums charged, ensuring the liability reflects a neutral position (i.e., no immediate profit).

3.18. Over time, the CSM is amortized into profit or loss as insurance services are provided, usually based on coverage units that reflect the expected pattern of service delivery.

3.19. Essentially, the CSM acts as a profit deferral mechanism, smoothing earnings recognition over the life of the contract.

Risk Adjustment (RA)

3.20. The **Risk Adjustment ("RA")** is the compensation an entity requires for bearing the uncertainty about the amount and timing of the cash flows that arise from non-financial risk as the entity fulfils insurance contracts.

3.21. In line with the Standard, several key non-financial risks are included in the risk adjustment calculation to capture the uncertainties inherent in the insurance contracts.

3.22. These non-financial risks mainly relate to:

a. **Mortality risk** – uncertainty about when policyholders will die.
b. **Morbidity risk** – uncertainty in illness, disability, or health-related claims under life & health riders.
c. **Lapse / Persistency risk** – uncertainty about whether policyholders will continue paying premiums or surrender policies early. Early lapses may cause losses if acquisition costs haven't been recovered.
d. **Expense risk** – uncertainty around future maintenance and claims handling costs.

3.23. A copy of the current guidelines as prescribed by the IRA are provided in the Appendix.

LRC Computations

3.24. LRC is calculated using the General Measurement Model (GMM) approach because the coverage of the contracts are often greater than one year as per the Standard.

3.25. The closing LRC at the end of each reporting period, e.g., as at 31 December 2024, is computed as follows:

3.26. If the present value of expected outflows exceeds expected inflows, that is, the contract is loss-making from inception or becomes loss-making later, the CSM cannot absorb this negative amount. In such cases, the excess is recognized immediately as a **Loss Component (LC)** in the LRC.

3.27. Ensures **onerous contracts** (unprofitable ones) are identified and accounted for and provides transparency on whether an insurer is writing contracts that are loss-making.

Liability for Incurred Claims (LIC)

3.28. For ordinary Life valuations, LIC is the outstanding liabilities which include the claims incurred and reported prior to the valuation date but are yet to be settled.

4 VALUATION WORKING FILE GUIDE

4.11. LRC Valuation for Ordinary Life business is normally carried out in the **IFRS 17 GMM Model**, which includes the following tabs;

Step by step guide to the working file

Introduction & Assumptions

4.12. In the ***Introduction*** tab, update the **valuation date**.

4.13. In the ***Assumptions*** tab, update the following as at the valuation Date;

i. Policy Admin fee

4.14. This is a charge applied to a policy to cover the insurer's administrative costs.

ii. Expense inflation

4.15. This is the assumed annual increase in expenses over time, reflecting the impact of general inflation or rising costs of doing business. *For example, if administration costs are expected to grow at 5% per year, an expense inflation rate of 5% would be applied.*

iii. Include future bonuses

4.16. This determines whether the calculations should take into account potential bonuses that may be declared in the future. Selecting **"Yes"** means the valuation includes assumptions for such bonuses, while **"No"** excludes them.

Source Data

4.17. This is the tab where the cleaned valuation data from the data clean up working file is uploaded.

i. Paste the valuation data in the respective columns.

ii. Update the "worksheet last row" reference to reflect the last row of the valuation dataset as shown below.

Worksheet - Last Row		**414.00**

iii. Click on the "Start" button as shown below to generate the result summaries.

!! *This step should only be performed once all required information across every tab has been updated* !!.

Results

4.18. Delete this tab before clicking the Start button in Source Data above. Otherwise, an error will appear prompting you to delete it first.

4.19. The results generated will include the components discussed in Chapter 2.

GMM Model

4.20. The 'GMM Model' tab is where all simulations are executed, and results are generated. It consolidates inputs from all other tabs and produces the result summaries. Do not make any changes to this tab.

Commissions

4.21. In this tab, paste the commission factors, (Year 1 commission, Year 2 commission…etc.) for each product as provided by the client.

Bonus

4.22. In this tab, paste the bonus rates for each product as provided by the client.

Yield Curve

4.23. Update the yield curve to reflect the NSE yield rates as at the end of the valuation period you are working on. *For example, the NSE yield curve as at 31st December 2024 represents the yield rates for year 2024.*

Risk Margin Factors

4.24. In this tab, paste the Risk Margin factors for each product as at the valuation date.

Product Information and Partial Payments

Product information Tab

4.25. The **Product Information** tab contains key features of the client's products under valuation, including but not limited to;

i. the product type
ii. commissions and expenses
iii. cover amounts
iv. risks covered
v. partial maturities available
vi. benefit terms available.

Partial Payments Tab

4.26. Based on the products information listed above, different products will have different codes. An Example is as shown below;

Partial Payment Table					
A1	B1	B2	C1	C2	D

4.27. The following should be updated;

1. The Partial Payments Rules

4.28. In this table, update the different product benefit payments based on the product features from the Product Information Tab discussed above. These products are only those that have partial payments.

Partial Payment Rules

	A1	B1	C1	D
1	20%	10%	20%	15%
2	20%	10%	25%	25%
3	30%	10%	100%	100%
4	30%	10%		
5		100%		
6				
7				

4.29. In the table above, update the product benefit payments according to the features outlined in the Product Information tab above. Only products with partial payments are included in this tab. For example;

"Product A1" is a product with the following partial maturity benefits; 4 years to maturity 20% of SA, 3 years to maturity 20% of SA, 2 years to maturity 30% of SA and 1 year to maturity 30% of SA.

2. The Payment Indicators

4.30. In this section of the tab, update the payment details as specified for each product. For example, for Product A1, the payment corresponds to the last four years to maturity (as per the product description). The indicator for this product will therefore be "Last 4," as shown below;

Last 4	Last 5	Last 6	4th &7th	Thirdly	6th & 12th
6	5	4		4	
Payable Indicator					
A1	B1	B2	C1	C2	D

Product Mapping

4.31. In this tab, update the following;

a. Products – as provided.

b. Commission Factors – from the Commissions tab.

c. Mortality Table – as applicable.

d. Profits – indicate whether the product is "With Profits" or "Without Profits".

e. Survival Payments – include partial payments from the Partial Payments tab, as well as "Last Year" payments for products without partial maturity benefits.

f. Premium Payment Mode – indicate whether regular or otherwise.

g. Per Policy Cost – as indicated in the Assumptions tab.

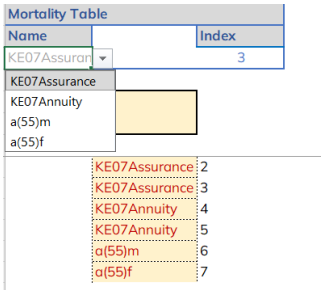
h. Expenses (% of Premium and % of Claims) – if applicable.

i. Payment Guarantee – indicate Yes or No depending on whether the product has a payment guarantee (as described in the product key features).

Mortality Tables and the nPx tabs

4.32. The ***Mortality Tables tab*** contains the KE 2007-2010 Assurance and Annuity tables, and the a(55) UK Tables, each provided separately for males and females.

4.33. In the ***nPx tab***, select the Mortality table to use from these options, as illustrated below;



5 APPENDICES

Appendix 1: Assumptions

5.11. The IRA has issued guidance on assumptions to be used when valuing liabilities. The following table summarizes the guidance given by IRA and the actual application of this for this valuation.

Key Assumption	IRA guidance
Mortality & Longevity	The KE07:10 base mortality rates will be used, with an appropriate adjustment to reflect the company's own experience.
Morbidity/Disability Rates	Where no reliable assumptions are available, the assumptions used can be based on the insurer's own experience, industry study or other relevant studies.
Withdrawals	The lapse rate should reflect the expected experience of both existing and potential policyholders, and the Actuary should perform a lapse investigation using the insurers past data. If a sufficient volume of relevant experience is not available, then experience from similar contracts and/or industrywide data (if available) may be used.
Expenses and Commissions	Actuarial judgement - The Actuary should make an allowance for expenses to include acquisition costs, underwriting & administration costs, investment costs, claim settlement costs and any other future expenses.
Investment Return	The Actuary should take into account the following regarding the future investment return assumption: a) the extent of any investment guarantees included; i. nature of contract; non-profit vs. with-profit ii. level of investment guarantee b) the size of the reserve built up under the contract; i. type of contract ii. frequency of premium payment
	The assumed rate of future investment return will depend on the mix of the assets held to match the liabilities
	Allowance should be made for any changes in the future economic environment
Other Assumptions	If a sufficient volume of relevant experience is not available, then experience from similar contracts and/or industrywide data *(if available)* may be used.

Appendix 2: Prescribed Risk Margins

Assumption	Prescribed Margin as % of the Base Assumption
Mortality	10% increase in mortality for death assurances
Longevity	10% decrease in mortality for life assurances and annuities
Morbidity/Disability	10% increase in inception rates 5% decrease in recovery rates
Lapses	25% *(increase or decrease in lapse rate depending on which alternative gives rise to an increase in the liability of the policy concerned)*
Interest rate (Investment Return)	20% decrease
Surrenders	10% *(increase or decrease in surrender rate depending on which alternative gives rise to an increase in the liability of the policy concerned)*
Expenses	10% increase
Expense inflation	10% increase *(of the estimated escalation rate)*

6 APPROVAL

Thus, adopted and signed by the management and Chief Actuary of KAFS in January 2026.

APPROVED BY

Ezekiel Macharia

Chief Actuary & Managing Director – Kenbright Actuarial and Financial Services Limited

Date: January 2026

Signed:

kenbright

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